

# Target-Visibility Control Barrier Function

A minimal-intervention lean projection on the real camera plane, with a self-releasing descent governor

Soft-Precise-Landing

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*Purpose.* Keep the marker’s measured centre inside the *real* camera image plane, with a fixed buffer from the field-of-view edge, by making the smallest change to the desired inertial acceleration  ${}^{\mathcal{I}}\mathbf{a}_d$  an outer loop produces. For the cross marker the visibility target is a *single point*, the crossed-lines intersection — no multi-corner containment, no marker-size term.

The command influences that point through two channels on different timescales, so it is conditioned in two tiers rather than one problem:

- **Attitude (lean).** A body tilt rotates the field of view, so the projected centre shifts *within one control step*, and depth-independently. This channel carries a hard barrier (§3) — the CBF.
- **Descent rate.**  $|\tilde{\mathbf{r}}| \approx |\text{offset}|/Z$ , which grows as the altitude  $Z$  shrinks. A change in the vertical command reaches the centre only two integrators later, so descent is treated as a *soft, predictive* influence: a governor (§5) that slows the descent when the centre is predicted to reach the buffer before the lateral loop can re-centre it.

## Notation

| Symbol                                                  | Meaning                                                                                                                                                                                   |
|---------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\tilde{\mathbf{r}}, \tilde{r}_k$                       | <i>measured</i> camera-frame marker centre, tangent units (px/ $f$ ), per axis $k \in \{1, 2\}$                                                                                           |
| $\mathbf{c}_{\text{next}}$                              | one-step predicted centre, (2)                                                                                                                                                            |
| $\mathcal{R}, f$                                        | sensor resolution and focal length (px)                                                                                                                                                   |
| $b$                                                     | FoV-edge buffer fraction, CBF_BUFFER_FRAC = 0.15;<br>$\varphi = \frac{\mathcal{R}}{2f}(1 - b)$                                                                                            |
| $L_\omega(\tilde{\mathbf{r}})$                          | rotational IBVS interaction matrix $\begin{bmatrix} xy & -(1+x^2) \\ 1+y^2 & -xy \end{bmatrix}$ at $(x, y) = \tilde{\mathbf{r}}$<br>— <b>contains no depth</b>                            |
| $M = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$     | fixed 90° lean→rotation-axis map: a lean along $\mathbf{y}$ is a body rotation about $M\mathbf{y}$                                                                                        |
| $L_e = -(L_\omega M)$                                   | effective Jacobian $\partial \mathbf{c}_{\text{next}} / \partial \mathbf{y}$                                                                                                              |
| $P = R_z(90^\circ)R_z(-\psi_b)$                         | inertial $x$ - $y \rightarrow$ image axes ( $\psi_b$ = body yaw; the 90° is the camera mount offset)                                                                                      |
| ${}^{\mathcal{I}}\mathbf{a}_d$                          | desired inertial <i>specific thrust</i> (thrust/mass), NED. Hover = $-g\mathbf{e}_3$ , so $\ {}^{\mathcal{I}}\mathbf{a}_d\ $ is the thrust magnitude per unit mass                        |
| $a_z =  {}^{\mathcal{I}}a_{d,z} $                       | vertical component magnitude — an <b>input</b> , never a decision variable                                                                                                                |
| $\mathbf{y} = P({}^{\mathcal{I}}\mathbf{a}_{d,xy}/a_z)$ | commanded lean, image axes; $\mathbf{y}_d$ from the unconstrained ${}^{\mathcal{I}}\mathbf{a}_d^0$ ;<br>$\mathbf{y}_{\text{curr}} = P(-R_{[:2,3]}/R_{33})$ from the realized attitude $R$ |
| $\mathbf{d}, \tau$                                      | measured translational drift rate of the centre (tangent/s, gyro-stripped); look-ahead horizon (s)                                                                                        |
| $A_{\text{cap}}, g$                                     | vehicle maximum thrust-to-mass ratio; gravitational acceleration                                                                                                                          |
| $g_z \in [g_{\min}, 1]$                                 | descent-governor scale (§5); $g_{\min} = \text{CBF\_GMIN} = 0.2$ , $T_{\text{react}} = \text{CBF\_TREAT} = 1.5\text{ s}$                                                                  |

## 1 Barrier and safe set

Per image axis  $k$ , define

$$\boxed{h_k(\tilde{\mathbf{r}}) = \varphi_k - |\tilde{r}_k|}, \quad \mathcal{S} = \{\tilde{\mathbf{r}} : h_k(\tilde{\mathbf{r}}) \geq 0, \ k = 1, 2\}. \quad (1)$$

$\mathcal{S}$  is the closed rectangle  $[-\varphi_1, \varphi_1] \times [-\varphi_2, \varphi_2]$ : the set of centre positions that lie on the sensor with at least the fractional margin  $b$  to each edge.  $\tilde{\mathbf{r}}$  is the *measured* centre — nothing is de-rotated onto a virtual frame.  $\varphi$  is a fixed camera constant. The buffer  $b$  is the robustness margin of the barrier: it absorbs, in one number, the intended visual margin, the one-step linearization residual of (2), and residual inner-loop attitude-tracking lag, so that a barrier value computed from the linear model is a lower bound on the true one to within the design regime of §2.

## 2 Feature response to the acceleration command

A camera rotation moves the centre by the rotational part of the IBVS interaction matrix, a standard *depth-independent* object evaluated at the measured point:  $\dot{\tilde{\mathbf{r}}} = L_\omega \boldsymbol{\omega}_{\text{rp}} + \mathbf{d}$ , with  $\boldsymbol{\omega}_{\text{rp}}$  the body roll/pitch rate.

The quantity a lateral-acceleration command sets is not a rotation rate but the underactuated quadrotor’s *lean* — the ratio of lateral to vertical thrust, expressed in image axes,  $\mathbf{y} = P(\mathcal{I} \mathbf{a}_{\text{d},xy}/a_z)$ . Producing a steady lean along  $\mathbf{y}$  is a body rotation about the perpendicular axis,  $\boldsymbol{\omega}_{\text{rp}} = M\mathbf{y}$ ; a world-fixed point’s image moves opposite to that rotation, so

$$\mathbf{c}_{\text{next}}(\mathbf{y}) = \tilde{\mathbf{r}} + L_e(\mathbf{y} - \mathbf{y}_{\text{curr}}) + \tau \mathbf{d}, \quad L_e = -(L_\omega M). \quad (2)$$

The exact map of  $\tilde{\mathbf{r}}$  under a finite lean change is the rotation homography of its ray; (2) is its first-order term about the current lean, and the sign of  $L_e$  is verified against an independent pinhole-plus-attitude camera model (`tools/validate_visibility_projection.py` check 1).  $\mathbf{y}_{\text{curr}}$  is read from the realized attitude, so it needs no rotation from acceleration units. The  $\tau \mathbf{d}$  term extrapolates measured *target* motion over the horizon  $\tau$ ; it is zero for a stationary target.

**Validity envelope.**  $L_e$  is evaluated at the measured  $\tilde{\mathbf{r}}$  and held fixed for the solve. The model error of (2) is under 8% for a per-cycle lean *change*  $\|\mathbf{y} - \mathbf{y}_{\text{curr}}\| \lesssim 15^\circ$ . At the 50 Hz control rate with a smooth outer loop this is the operating regime; a single cycle never demands a large lean change. Empirically (4000 randomized geometries per bin): 0 true-sensor exits at  $\leq 15^\circ/\text{cycle}$ ; a 0.35% rate at  $\sim 24^\circ/\text{cycle}$  with worst overshoot +0.07 tangent, well inside  $b\mathcal{R}/(2f)$ .

## 3 Tier 1 — the visibility CBF

Each control step, given the outer loop’s unconstrained command  $\mathcal{I} \mathbf{a}_{\text{d}}^0$  (hence  $\mathbf{y}_{\text{d}}$ ) and the measurement  $(\tilde{\mathbf{r}}, \mathbf{d}, R)$ ,

$$\boxed{\mathbf{y}^* = \arg \min_{\mathbf{y} \in \mathbb{R}^2} \frac{1}{2} \|\mathbf{y} - \mathbf{y}_{\text{d}}\|^2 \quad \text{s.t.} \quad |\mathbf{c}_{\text{next}}(\mathbf{y})|_k \leq \varphi_k, \quad k = 1, 2} \quad (3)$$

after which  $\mathcal{I} \mathbf{a}_{\text{d},xy}^* = a_z P^\top \mathbf{y}^*$  and the vertical command  $\mathcal{I} a_{\text{d},z}$  is passed through unchanged. (3) is a convex quadratic program: the objective is strictly convex and each constraint  $|\mathbf{c}_{\text{next}}|_k \leq \varphi_k$  is a pair of half-planes affine in  $\mathbf{y}$  (its gradient is the fixed row  $L_{e,k}$ ), so the feasible set  $\mathcal{Y}$  is a convex polyhedron.

**This is a discrete-time control barrier function.** The constraint of (3) is exactly  $h_k(\mathbf{c}_{\text{next}}(\mathbf{y})) \geq 0$  for both  $k$ : it requires the *next-step* barrier value to be non-negative. This is the discrete-time CBF condition  $h(\mathbf{x}_{k+1}) \geq \gamma h(\mathbf{x}_k)$  at  $\gamma = 0$  (hard invariance), with a  $\tau \mathbf{d}$  look-ahead so the invariance also anticipates exogenous target motion.

**Proposition 1** (Forward invariance). *Let  $\hat{\mathbf{c}}_k = \mathbf{c}_{\text{next}}$  evaluated with the realized attitude and drift of step  $k$ , and suppose (i) (3) is feasible at every step, and (ii) the realized centre at step  $k+1$  equals  $\hat{\mathbf{c}}_k$  to within  $b\mathcal{R}/(2f)$  (the model regime of §2). If  $\tilde{\mathbf{r}}_0 \in \mathcal{S}$  then  $\tilde{\mathbf{r}}_k \in \mathcal{S}$  for all  $k$ .*

*Proof.* By induction. If  $\tilde{\mathbf{r}}_k \in \mathcal{S}$ , feasibility of (3) gives a  $\mathbf{y}^*$  with  $|\hat{\mathbf{c}}_k|_k \leq \varphi_k$ , i.e.  $\hat{\mathbf{c}}_k \in \mathcal{S}$ . By (ii) the realized  $\tilde{\mathbf{r}}_{k+1}$  differs from  $\hat{\mathbf{c}}_k$  by at most the buffer margin, so  $|\tilde{\mathbf{r}}_{k+1,k}| \leq \varphi_k + b\mathcal{R}/(2f) = \mathcal{R}/(2f)$ :

the centre is still on the sensor.  $\square$  The buffer thus converts the visual margin into a certified tolerance for the one-step linearization and attitude-tracking residual. Feasibility of (3) holds whenever  $\varphi_k > 0$  for both  $k$  (the marker does not subtend the full buffered FoV); §4 discusses the interaction with the deliverability caps.

**Proposition 2** (Pointwise-optimal, minimal intervention). *If  $\mathbf{y}_d \in \mathcal{Y}$  then  $\mathbf{y}^* = \mathbf{y}_d$ ; otherwise  $\mathbf{y}^*$  is the Euclidean projection of  $\mathbf{y}_d$  onto  $\mathcal{Y}$ , unique by strict convexity.*

The CBF is inert whenever the outer command is already visibility-safe, and otherwise perturbs it by the least  $\|\mathbf{y} - \mathbf{y}_d\|$  that restores safety. In flight on a well-behaved approach the projection is a pass-through for essentially every frame.

**Proposition 3** (No strangling of recovery). *Leaning toward the marker, or tangentially to it, does not increase  $|\mathbf{c}_{next}|_k$  on the binding axis; such a command is left unchanged.*

*Sketch.* A command that reduces  $|\tilde{r}_k|$  makes constraint  $k$  slack; the projection in (3) touches only rows that are active (violated), so a slack row’s command component passes through. Concretely the alternating projection below adjusts  $\mathbf{y}$  only along  $L_{e,k}$  for a  $k$  with  $|\hat{c}_k| > \varphi_k$ ; a re-centring  $\mathbf{y}_d$  has  $|\hat{c}_k| \leq \varphi_k$  and is not adjusted.

**Solver.** (3) is solved by a few sweeps of cyclic projection of  $\mathbf{y}_d$  onto the (at most two) violated half-planes. Because the two constraint normals  $L_{e,1}, L_{e,2}$  are non-parallel for any  $\tilde{\mathbf{r}}$ , cyclic projection converges geometrically to a point of  $\mathcal{Y}$ ; when at most one row is ever active — the common case, since the marker rarely approaches two edges at once — a single projection is the exact minimum-norm solution. Three to four sweeps suffice in practice.

## 4 Deliverability

Thrust deliverability is enforced *after* (3), not folded into it, which is what keeps  $a_z$  an input and the CBF a plain feasibility projection. Two bounds are applied in order:

1. **Lean cap:**  $\|\mathbf{y}^*\| \leq \arccos(a_z/A_{\text{cap}})$  — the true deliverable tilt at the actual  $a_z$  (which equals  $\theta_{\text{cap}}$  only at hover, and correctly loosens below hover / tightens above it). If it binds,  $\mathbf{y}^*$  is scaled radially toward the origin and  ${}^{\mathcal{I}}\mathbf{a}_{d,xy}^*$  rescaled to match.
2. **Thrust cap:**  $\|{}^{\mathcal{I}}\mathbf{a}_d\| \leq A_{\text{cap}}$  on the full vector.

Radial scaling of  $\mathbf{y}^*$  toward  $\mathbf{0}$  moves  $\mathbf{c}_{next}$  toward  $\tilde{\mathbf{r}} + L_e(-\mathbf{y}_{\text{curr}})$ ; near level flight ( $\mathbf{y}_{\text{curr}}$  small) this is close to the measured  $\tilde{\mathbf{r}}$ , which is in  $\mathcal{S}$  whenever the CBF is active at all. The caps therefore tighten toward the in-frame measured centre rather than away from it; they do not, in the operating regime, invalidate Proposition 1.

## 5 Tier 2 — the descent-rate governor

A change in  $a_z$  does not move  $\mathbf{c}_{next}$  this cycle, so it lies outside the CBF. It is handled as a soft governor. Let, per axis,

$$\text{rem}_k = \varphi_k - |\tilde{r}_k|, \quad \text{close}_k = \max(\text{sgn}(\tilde{r}_k) \dot{\tilde{r}}_k, 0), \quad t_{\text{edge}} = \min_k \frac{\text{rem}_k}{\text{close}_k}, \quad (4)$$

the earliest predicted time for the centre to reach the buffer at its measured outward image speed ( $t_{\text{edge}} = \infty$  if it is not moving outward). Then

$$\boxed{g_z = g_{\min} + (1 - g_{\min}) \text{clip}\left(\frac{t_{\text{edge}}}{T_{\text{react}}}, 0, 1\right), \quad \mathcal{I}a_{d,z}^{\star\star} = (\mathcal{I}a_{d,z}^{\star} + g)g_z - g,} \quad (5)$$

applied only while a descent is commanded ( $\mathcal{I}a_{d,z}^{\star} + g > 0$ ) and low-passed so  $a_z$  does not step. Only the *downward* part of the vertical command is scaled;  $g_{\min} > 0$  keeps a minimum descent and the map cannot command a climb.

**Self-releasing.**  $g_z$  is monotone increasing in  $t_{\text{edge}}$ , and  $t_{\text{edge}}$  increases as  $|\tilde{\mathbf{r}}|$  decreases (larger rem, smaller close). Slowing the descent gives the lateral loop more cycles at a given altitude to reduce  $|\tilde{\mathbf{r}}|$ ; as it does,  $t_{\text{edge}} \rightarrow \infty$  and  $g_z \rightarrow 1$ . The trigger is an external *measured* quantity ( $|\tilde{\mathbf{r}}|$  and its rate), so the governor cannot self-sustain on a quantity it generates, and it is applied after Tier 1 so it never enters the lean solve. It is not a barrier and makes no invariance claim; its role is to widen the time window in which Proposition 1’s feasibility assumption holds near touchdown.

## 6 Assumptions

**Assumption 1.** *The camera-mount rotation  $P$  and the coupling  $L_e = -(L_\omega M)$ , with its sign, are calibrated (validated against an independent pinhole model).*

**Assumption 2.** *The conditioned command is realized: the inner attitude loop tracks the attitude built from  $\mathbf{y}^{\star}$  on a timescale fast relative to the outer loop, with residual lag inside the buffer  $b$ .*

**Assumption 3.**  *$\mathbf{d}$  is the translation part of the observed centre velocity, gyro-stripped;  $\tau = 0$  for a stationary target.*

**Assumption 4.** *The per-cycle lean change stays within the first-order regime of §2 ( $\lesssim 15^\circ$ ), which the 50 Hz rate and a smooth outer loop provide.*

**Assumption 5.**  *$A_{\text{cap}}$  and  $g$  are fixed vehicle/physical constants, used directly.*

## 7 Scale-freeness and depth-freeness

Every quantity in (1), (2), (3), (4)–(5) is one of: a fixed camera intrinsic ( $\mathcal{R}/(2f)$ ,  $b$ ), an attitude ( $R$ ,  $\psi_b$ ), an image-space measurement ( $\tilde{\mathbf{r}}$ ,  $\dot{\tilde{\mathbf{r}}}$ ,  $\mathbf{d}$ ), a ratio or pass-through of the outer loop’s own command ( $\mathbf{y}$ ,  $a_z$ ), a fixed physical constant ( $A_{\text{cap}}$ ,  $g$ ), or a fixed time constant ( $\tau$ ,  $T_{\text{react}}$ ).  $L_\omega$  is depth-independent by construction.  $t_{\text{edge}}$  is an image distance over an image speed — seconds, dimensionless in scale. No target range, altitude, or metric velocity appears anywhere. The formulation is manifestly scale- and depth-free.